

BME 383J/CSE 397, AI Methods in Medical Imaging

UNIQUE NUMBERS: 168840, 70444

Semester: Fall 2026

MEETING TIME AND LOCATION: Tue/Th: 12:30PM - 2:00PM, GDC 5.304 (in-person)

INSTRUCTOR: Umberto Villa

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Office hours: Tue: 2-3pm, Wed: 10-11am
and by appointment

COURSE DESCRIPTION

The course provides a comprehensive understanding of the fundamental principles of image formation and quality assessment, essential for the principled application of artificial intelligence (AI) in biomedical imaging. Imaging systems are analyzed using both a deterministic framework, employing linear operator theory, and a statistical framework, introducing stochastic models for objects and images. Methodologies for the objective assessment of image quality (OAIQ) will be introduced, which will address signal detection tasks, receiver operator characteristic (ROC) analysis, numerical observers, and estimation theory. Students will be assessed based on participation (required readings and discussions, 10%); three take-home assignments (20% each); and a final group project (written report and oral presentation; 30%). Take-home assignments and the group project will allow students will include some Python programming to put the concepts learned in class into practice and gets hand-on experience with the methods introduced in the course.

UNIVERSITY CATALOG COURSE DESCRIPTION

Explore fundamental principles of image formation and quality assessment, essential for the principled application of AI in biomedical imaging. Introduce methodologies for the objective assessment of image quality (OAIQ) to address signal detection tasks, receiver operator characteristic (ROC) analysis, numerical observers, and estimation theory.

PRE-REQUISITES FOR THE COURSE

Graduate standing and consent of instructor. Familiarity with linear algebra and numerical optimization concepts is expected, as are basic programming skills in Python. No prior experience with machine learning or medical image reconstruction is assumed.

LEARNING OUTCOMES

1. Apply linear operator theory to describe imaging systems, understand the concept of null-space and ill-posedness.
2. Formulate image reconstruction as model-based optimization and statistical inference problems.
3. Apply large-scale numerical optimization techniques and AI-based methods to image reconstruction problems.
4. Apply objective measures of image quality for the assessment and optimization of imaging system design and image reconstruction algorithms.

HOW WILL YOU LEARN?

COURSE CANVAS SITE

<https://utexas.instructure.com/courses/1457970>

TEACHING MODALITY INFORMATION

The course modality is in-person. In-person attendance is expected; however, lecture recordings will be posted on Canvas for future reference.

COMMUNICATION

The course Canvas site can be found at utexas.instructure.com. Please email me through Canvas. You are responsible for ensuring that the primary email address you have recorded with the university is the one you will check for course communications because that is the email address that Canvas uses.

ASKING FOR HELP

Students are encouraged to ask questions before, during, after class. All questions are welcome, and none of them are an interruption. If something in a lecture doesn't land, raise your hand right then; the odds are high that several other people are stuck on the same thing and are hoping someone else will speak first. Be that person. Questions that feel too basic to ask out loud are usually the ones worth asking, because they tend to be about the assumption everything else rests on. Outside of class, come to office hours or email me through Canvas, and I will also meet by appointment if those times don't work.

Please ask **early**. This material builds on itself, so a small confusion in week 3 becomes a large one by week 8. A ten-minute conversation the day something stops making sense will save you hours later. If you are stuck on an assignment, get in touch before the deadline rather than after it.

ACADEMIC ACCOMMODATIONS

Students with disabilities may request appropriate academic accommodations from the Division of Student Affairs, Disability and Access (D&A) – see the [University Policies and Resources for Students](#) Canvas page or visit the [D&A website](#).

COURSE REQUIREMENTS AND GRADING

REQUIRED MATERIALS

All required materials and lecture notes will be posted on Canvas.

RECOMMENDED MATERIALS

Image science and objective assessment of image quality:

Harrison H. Barrett and Kyle J. Myers. *Foundations of Image Science*. Wiley, Hoboken, New Jersey, 2004. ISBN: 0-471-15300-1.

Draft chapters from the book are freely available from Dr. Meredith Kupinski's website at the University of Arizona:

<https://sites.arizona.edu/pol-lab/resources/>

A landmark book providing a comprehensive description of all aspects of imaging science.

Model-based and statistical image reconstruction:

Charles A. Bouman. *Foundations of Computational Imaging: A Model-Based Approach*. SIAM, Philadelphia, 2022. ISBN: 978-1-611977-12-7.

Freely accessible from publisher website via UT subscription (must be on campus or VPN):

<https://epubs.siam.org/doi/book/10.1137/1.9781611977134>

A unified treatment of Bayesian and regularized reconstruction, image priors, majorization, constrained optimization, ADMM, and plug-and-play methods.

Numerical optimization

Amir Beck. *First-Order Methods in Optimization*. MOS-SIAM Series on Optimization, SIAM, Philadelphia, 2017. ISBN: 978-1-611974-98-0.

Freely accessible from publisher website via UT subscription (must be on campus or VPN):

<https://epubs.siam.org/doi/book/10.1137/1.9781611974997>

A self-contained treatment of the first-order methods used in this course, including proximal gradient methods, FISTA, and ADMM.

Deep learning for image reconstruction

Reinhard Heckel. Deep Learning for Computational Imaging. Oxford University Press, Oxford, 2025. ISBN: 978-0-19-894717-2 (hardback), 978-0-19-894718-9 (paperback).

Freely accessible from publisher website via UT subscription (must be on campus or VPN):

<https://academic.oup.com/book/60001>

A new book covering modern deep learning-based approaches to image reconstruction

SHARING OF COURSE MATERIALS IS PROHIBITED

No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class without my explicit, my written permission. Unauthorized sharing of materials may facilitate cheating. The University is aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to [Student Conduct and Academic Integrity](#) in the Office of the Dean of Students. These reports can result in initiation of the student conduct process and include charge(s) for academic misconduct, potentially resulting in sanctions, including a grade impact.

REQUIRED DEVICES

A computer with a working python installation is required for the take-home assignments and final group project.

CONFIDENTIALITY OF CLASS RECORDINGS

[HOP 2-9970](#) prohibits students from recording class instruction (audio or video) unless a student obtains the instructor's permission or Disability & Access has approved audio recording as an accommodation.

Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

GETTING HELP WITH TECHNOLOGY

To learn more about available technologies for students or if you require technical assistance, please visit the [Technology for Students](#) website. Students needing help with technology in this course should contact the [Support Center](#) or [Sysnet](#) (CSEM students using Oden Institute's computers only).

CLASSROOM EXPECTATIONS

Class attendance. Attendance is expected, but it will not be recorded. Students are encouraged to let the instructor know in advance if missing class because of conference travel or other circumstances. Recorded lectures will be posted on Canvas. **Attendance is strictly required for the final project presentations that will be held December 1st and December 3rd, 2026.**

Class participation. Students are encouraged to actively participate during class by asking questions and discussing the required readings.

Behavior expectations. My goal is a classroom where everyone feels safe learning: safe asking a basic question, safe being wrong out loud, safe disagreeing. That requires engaging with ideas rather than the people holding them, and treating your classmates' half-formed thoughts the way you'd want yours treated. Arrive on time, stay present, and in team work do your share. Formal ground rules for student conduct are in [Section 11-400](#) of the Institutional Rules in the *General Information Catalog*.

ARTIFICIAL INTELLIGENCE POLICY STATEMENT

In accordance with the University's Institutional Rules on Student Services and Activities, Chapter 11, students accept the responsibility to always uphold academic integrity and an honor code reflective of a scholarly community devoted to academic and personal success. All members of the University community are fully accountable and responsible for any output they produce as part of academic work. They are also responsible for following the guidance specified in the [Texas Statement on Academic Integrity](#) and avoiding prohibited uses of generative AI tools outlined in the Information Security Office's guidance on [Acceptable Use of Generative AI Tools](#).

Generative AI use is partially permitted for academic work in this course, provided that students 1) use AI [responsibly](#), 2) practice discernment and conduct meaningful human review of any output generated by AI, and 3) properly disclose use according to the disclosure policies (see the *Generative AI Use Disclosure Policies* document on Canvas). You will be informed by your instructor in writing as to the aspects of the assignments for which AI may be utilized and how it may be utilized. Using generative AI tools without authorization or failing to disclose generative AI use according to the disclosure policy in this course, even when AI use is permitted, may constitute academic misconduct under UT Austin's Institutional Rules and may be referred to Student Conduct and Academic Integrity in the Office of the Dean of Students for resolution.

ACADEMIC INTEGRITY EXPECTATIONS

All allegations of academic misconduct will be referred by faculty to the Office of the Dean of Students. Students who violate University rules on academic misconduct are subject to the student conduct process. A student found responsible for academic misconduct may be assigned both a status sanction and a grade impact for the course.

The grade impact could range from a zero on the assignment in question up to a failing grade in the course. A status sanction can include a written warning, probation, deferred suspension, suspension, or expulsion from the University. To learn more about academic integrity standards, tips for avoiding a potential academic misconduct violation, and the overall conduct process, please visit the Student Conduct and Academic Integrity website at <https://deanofstudents.utexas.edu/conduct/student-conduct-processes.php>

ASSIGNMENTS

The following table represents how you will demonstrate your learning and how we will assess the degree to which you have done so.

Assignments	Points Possible	Percent of Total Grade
1. Participation (required readings and discussions)	10	10%
2. Take-home Assignment 1 (Due: September 24 th , 2026)	20 + 2*	20%
3. Take-home Assignment 2 (Due: October 22 nd , 2026)	20 + 2*	20%
4. Take-home Assignment 3 (Due: November 19 th)	20 + 2*	20%
5. Group Final Project Presentations (December 1 st & 3 rd , 2026)	10**	10%
6. Group Final Project Report (Due: December 7 th , 2026)	20**	20%
Total	100 + 6*	100%

* Each take-home assignment includes 2 buffer points (10% of the assignment grade).

** All students in each group will receive the same grade in the group final project and presentation.

RELIGIOUS HOLY DAYS

By [UT Austin policy](#), you must notify me of your pending absence for a religious holy day as far in advance as possible of the date of observance. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence. For questions regarding religious accommodations, please contact the [Office of the Dean of Students](#).

GRADING POLICY

Students will be assessed based on participation (required readings and discussion; 10%); three take-home assignments (20% each); and one final group project (written report and oral presentation; 30%).

MEANS OF EVALUATION AND ASSIGNMENT OF CLASS GRADES

Participation. The participation grade will be assessed holistically. It includes required readings as demonstrated by participating in classroom discussions, interactions with peers and instructor, and participating in Canvas surveys.

Take-home assignments. Take-home assignments include a mix of theoretical work and programming.

All take-home assignments are to be performed individually. Students are encouraged to discuss the assigned problems with each others and with the instructor, but may not share code or other work related to the take-home assignments. Students will submit their take-home assignments using Canvas as a single PDF document, this can be handwritten (on an iPad or scanned, but legible) or typed. When requested the document should include code snapshots and outputs (figures, tables) as detailed in each take-home assignment. As stated in the Generative AI Use Disclosure Policies every submission should end with a short AI Use Statement and reflection. This statement should describe, in the student's own words, what tool was used, for which tasks, how extensively, how the output was verified, and a short reflection (150-300 words) describe how the interaction with the AI tool enhanced the understanding of the class topics. If a student did not use AI tools for a specific assignment, the submission should still include a statement saying so. **A missing AI Use Statement is treated as an incomplete submission** and will not be graded until the student provides one. *I read AI Use Statements with interest, not suspicion: We are all learning how to best leverage this resource in a meaningful way. Honest, thoughtful disclosure is part of the work and is never penalized.*

Group final project. The group final project will be evaluated based on the quality of the written report and oral presentation. All students are expected to equally contribute to the project and all students in the same group will receive the same grade. The group final projects will consists of three phases.

1. **Project proposal** (Not graded, but nevertheless of utmost importance, October 9th). Students are encouraged to form small (2 or 3 people) groups early in the semester. Before starting work on the project, each group will submit a short project proposal detailing the scope of the project, the data/images that will be used, and the expected outcomes. This proposal will then be discussed with the instructor to ensure the project is relevant and feasible. Some example projects will be posted on Canvas; however, students are encouraged to propose their own project.
2. **Project presentations** (10% of total grade, December 1st & 3rd). Each group will present their work and engage in a Q&A with the other students and the instructor. It is expected for all team members to give a part of the oral presentation and to engage in the Q&A. **Students must attend both days, not just the one their group is presenting.**
3. **Written report** (20% of total grade, December 7th). Written reports should take the form of a short research article (about 20 pages including figures, tables, and references). They should feature an introduction that defines the target problem and provides context within the literature, a method section that describes the

implementation and assessment of the proposed methods, a result section, and finally a discussion and conclusion section. Each written report should also include both a **Contribution Statement** and **AI Use Statement** to indicate how students in each group have interacted among each other and with generative AI, respectively. These statements are required, and a final report without such statements will not be accepted.

LATE WORK AND MAKING UP MISSED WORK

Take-home assignments. Students should inform the instructor as soon as possible if they are going to miss a deadline due to conference travel, or other research or personal circumstances. Extensions will be granted on a case-by-case basis. Late take-home assignments will still be accepted with a 10% penalty if submitted within a week from the original (or extended deadline). Without prior permission from the instructor, a take-home assignment that is more than one week late will be considered missing and counted as 0.

Group final projects. The project presentation dates and due date for the written report are firm and no extension can be granted.

REGRADING REQUESTS

Regrading requests are accepted within a week from the date when grades are posted. Students are always welcome to ask questions and clarifications about their work at any time during the semester, but regrade requests will not be accepted past the deadline.

ABSENCES

Attendance is expected but it will not be recorded. **However, students are required to be present for the group project presentations that will be held on December 1st and 3rd, 2026.**

EXTRA CREDIT

No additional assignments/readings will be offered for extra credit. However, each take-home assignment will include 2 buffer points (10% of assignment grade).

GRADE BREAKS

Grade	Cutoff
A	92%
A-	88%
B+	85%
B	82%
B-	79%
C+	76%
C	73%
C-	70%
D+	67%
D	64%
D-	60%
F	<60%

COURSE OUTLINE

All instructions, assignments, readings, rubrics and essential information will be on the Canvas website at utexas.instructure.com. Check Canvas regularly. Changes to the schedule may be made at my discretion if circumstances require. I will announce any such changes in class and will also communicate them via a Canvas announcement. It is your responsibility to note these changes when announced, and I will do my best to ensure that you are notified of changes with as much advance notice as possible.

FINAL EXAM DATE AND TIME

There is not final exam for this course. Group project presentations will be held during the our last two class meetings.

UNIVERSITY POLICIES AND RESOURCES

For a list of important university policies and helpful resources that you may need as you engage with and navigate your courses and the university, see the [University Policies and Resources for Students](#) Canvas page. The page includes important safety information including the policy on carrying of handguns on campus, the language of the University Honor Code, Title IX legal requirements for Texas employees, and information about how to receive support through the office of Disability & Access, tutoring services, the Counseling and Mental Health Center, University Health Services, and other resources.